

B. Tech Degree VI Semester Examination April 2012

ME 604 HEAT AND MASS TRANSFER (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Derive the general differential equation of heat conduction? How this differential equation is subject to the given constraints called initial and boundary conditions of the problem?
- (b) What is called as Extended Surfaces? Why is the assumption of one-dimensional heat flow made in the analysis of Extended Surfaces?
- (c) Define Grashof Number. Explain its significance in natural convection heat transfer.
- (d) What is Reynold's Analogy? Describe the relation between the fluid friction and heat transfer.
- (e) State and explain the Wein-Displacement law. Show that the $E_{b\lambda}$ will be maximum when $\lambda_{\max} T = 2900 \mu k$.
- (f) What is radiation shape factor and why are they used? State and explain reciprocity theorem.
- (g) What is LMTD method of heat exchanger analysis? When is LMTD method most suitable for heat exchanger designer calculations?
- (h) How are heat exchangers classified? Sketch the temperature variations in parallel flow and counter flow heat exchangers.

PART B

(4 x 15 = 60)

- II. (a) In a cylindrical fuel rod of a nuclear reactor, the internal heat generation is given by the equation: (8)
- $$\dot{q} = \dot{q}_0 \left[1 - \left(\frac{r}{r_0} \right)^2 \right]$$
- Calculate the temperature drop from the center line to the surface of a 2.5cm O.D rod having a thermal conductivity of 20w/mk, if the rate of heat removal from the surface is 2.5 MW/m². (7)
- (b) Calculate the heat loss per sq.m of outside surface area for a heated sphere 15cm in diameter covered with 5cm layer of insulation (K = 0.0663W/mK). The inside and outside surface temperatures of the insulation are 315°C and 80°C respectively.

OR

- III. (a) A straight rectangular fin has a length of 35mm, thickness of 1.4mm. The thermal conductivity is 55 w/mk. The fin is exposed to a convection environment at 20°C and h = 500 w/m²k. Calculate the heat loss for a base temperature of 150°C. (8)
- (b) A copper wire of 40mm diameter carries 250A and has a resistance of $0.25 \times 10^{-4} \Omega$ /length. Surface temperature of copper wire is 250°C and the ambient air temperature is 10°C. If the thermal conductivity of the copper wire is 175 w/mk, calculate: (7)
- (i) Heat transfer coefficient
- (ii) Maximum Temperature

(P.T.O)

- IV. A wall 4m high by 5m wide is at 60°C and the surrounding air is at 20°C. Calculate the heat lost by natural convection neglecting end effects using the relation (15)

$$N_{Nu} = 0.51(N_{pr})^{1/2} (N_{pr} + 1)^{1/4} (N_{Gr})^{1/4}$$

Take $e = 1.09 \text{ kg} / \text{m}^3$, $\mu = 1.91 \times 10^{-5} \text{ NS} / \text{m}^2$

$$K = 0.063 \text{ KJ} / \text{mb}, K, N_{pr} = 0.714.$$

OR

- V. Using Buckingham's Π theorem, explain the dimensional analysis applied to Forced Convection and derive an expression $N_u = \phi(\text{Re}, P_r)$. (15)

- VI. The outlet header of high pressure steam super heater consists of a pipe ($E = 0.8$) of diameter 27.5cm. Its surface temperature is 500°C. Calculate the loss of heat per unit length by radiation if it is placed in an enclosure at 30°C. (15)

If the header is now enveloped in a steel screen of diameter 32.5 cm and emissivity 0.7 and the temperature of the screen is 240°C, find the reduction in heat loss by radiation due to provision of this screen.

OR

- VII. (a) What are called as Radiation Shields? (5)
 (b) Two perfectly black parallel planes of 1.2m x 1.2m are separated by a distance of 1.2m. One plane is maintained at 800K and the other at 500K. The planes are located in a large room whose walls are at 300K. What is the net heat transfer between them? (10)

- VIII. Water enters a counter flow, double pipe heat exchanger at 10°C, flowing at the rate of 1300kg/hr. It is heated by oil ($C_p = 2000 \text{ J/kgK}$) flowing at the rate of 550kg/hr from an inlet temperature of 94°C. For an area of 1 m^2 and an overall heat transfer coefficient of $1075 \text{ W/m}^2\text{K}$, determine the total heat transfer and the outlet temperatures of water and oil. (15)

OR

- IX. A counter flow double pipe heat exchanger using super heated steam is used to heat water at the rate of 10500kg/hr. The steam enters the heat exchanger at 180°C and leaves at 130°C. The inlet and exit temperature of water are 30°C and 80°C respectively. If the overall heat transfer coefficient from steam to water is $814 \text{ W/m}^2\text{K}$, calculate the heat transfer area. What would be the increase in area if the fluid flows were parallel? (15)